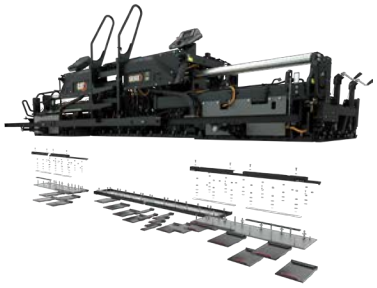
CAT SDX SCREED PLATE SYSTEM

The Cat SDX Screed Plate Technology is a field installed option that can help improve road quality with better consistency and higher reliability. This revolutionary design provides dynamic flow through the screed plates to reduce air voids and increase density.

The Cat SDX screed plate system is not be available in all regions. Please consult your Cat dealer for availability in your location.







INNOVATIVE MATERIAL FLOW

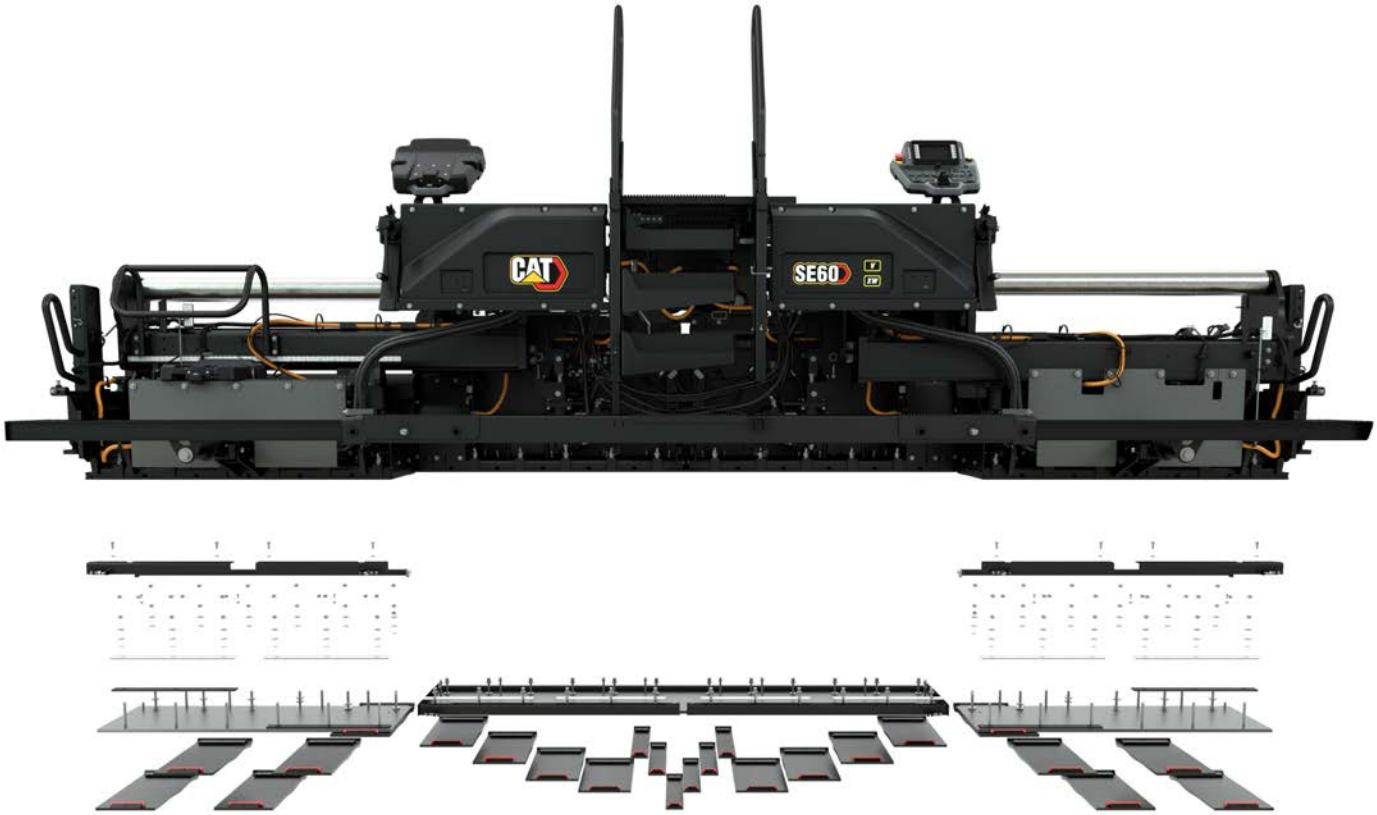
+ Angular surface provides a kneading action as it passes through the grooves
+ Tapering flat surface at the back of the plates provides a smooth finished surface
+ Confined mixing action manipulates aggregates to reduce air voids and increase density while creating a more uniform surface texture
+ Optimized aggregate structure promotes surface temperature uniformity for more consistent compaction
+ Enhanced structural integrity minimizes roller shoving for better smoothness

INCREASED DENSITY AND SMOOTHNESS

+ Kinetic mixing action helps increase the bond interface between the asphalt binder and aggregate
+ Mixing action of the angular plates is designed promote better thermal uniformity for improved compaction characteristics
+ Increased density can enable compaction to take place on a hotter surface shortening the compaction train
+ Increased stiffness can lead to better longitudinal edge profiles and higher joint densities due to less lateral movement of the mix when being compacted

FASTER SCREED PLATE CHANGES

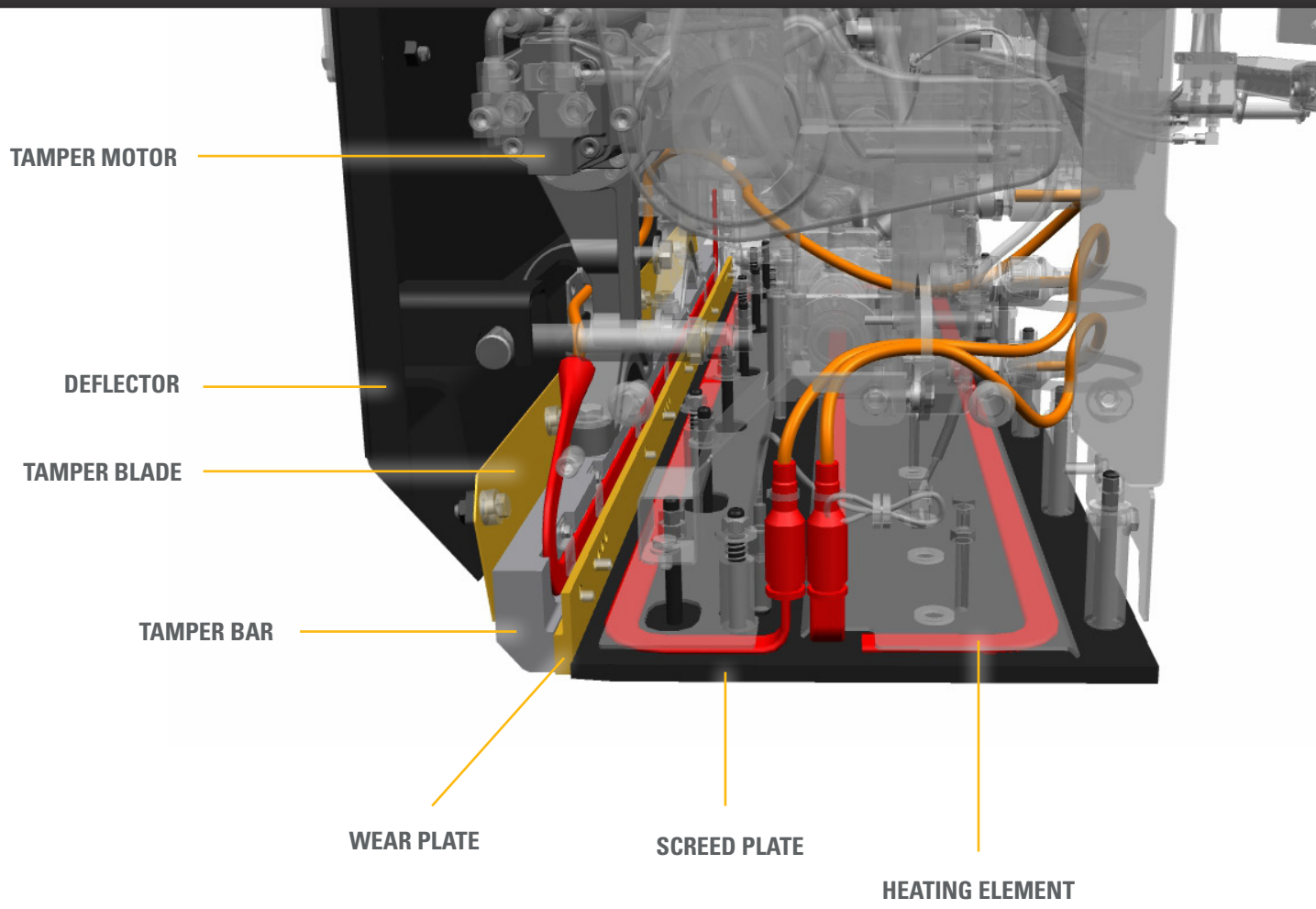
+ Hardware-free screed plate installation saves time and effort
+ Tapered blocks integrated into the screed plates fit over the beveled edge of the adapter plates and are held in place with high-temperature silicone retainers that maintain tension for reliable performance
+ One-time adapter plate install attaches to the screed frame utilizing the existing screed plate orientation
+ Heating elements attach to the adapter plates
+ Modular screed plates come in 300 mm (12") sections that weigh approximately 16 kg (35 lbs.). The plates are interchangeable across the screed width to help maximize wear.
+ The center screed section consists of 100 mm (4") plates that weigh approximately 5 kg (12 lbs.). These narrow plates allow for crown at the center of the screed.



CAT SDX SCREED PLATE SYSTEM KITS

DESCRIPTION	SCREED MODEL	SERIAL NUMBER PREFIX	PAVER MODEL	PART NUMBER
The Parts Kits come complete and include the aluminum adaptor plates, screed plates, nylon retainers, spacers, covers, and hardware. Available for order through your Cat dealer only. ***SDX may not be available in all regions. Please consult your Cat dealer for availability in your location.***				
SDX Screed Plate Kit*	SE60 V	JJ3	AP655, AP655F, AP1000, AP1000F, AP1055, AP1055F	642-9792
	SE60 V XW	JS9		640-1505
0.25 m Bolt–on Extension – Left side	SE60 V, SE60 V XW	JJ3, JS9	AP655, AP655F, AP1000, AP1000F, AP1055, AP1055F	643-0892
0.25 m Bolt–on Extension – Right side				643-0893
0.75 m Bolt–on Extension – Left side	SE60 V, SE60 V XW	JJ3, JS9	AP655, AP655F, AP1000, AP1000F, AP1055, AP1055F	643-1331
0.75 m Bolt–on Extension – Right side				643-1332

** This is a paper kit. Not available as a stocked kit but as a service repair option that may be ordered, at the consist level, through your Cat dealer to complete the repair or use SIS to identify part numbers for order.*





1. TAMPER BAR KIT Contains the tamper blades, wear plates, pins, bolts, nuts and washers for a basic tamper bar repair.

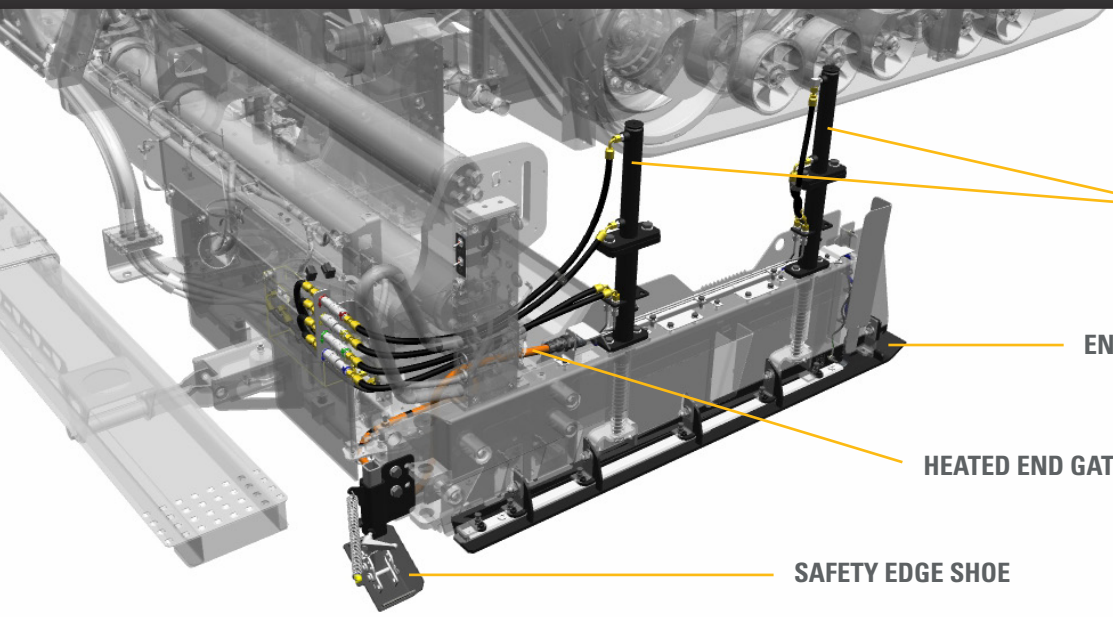


2. TAMPER BAR HEATING ELEMENT KIT This kit is designed to support electric heated screeds with a tamper bar. The kit contains the tamper bar heating element and required hardware. This kit does not include the tamper bar. Please refer to the Tamper Bar kit. **NOTE:** Apply thermal compound under every heat sensor and heating element during installation; **the compound is not included in this kit and must be ordered separately.** Dow Corning 340 heat sink compound is recommended to complete this repair. ***F-Series kit does not require thermal compound.***

SCREED TAMPER BAR KITS

SCREED MODEL	SERIAL NUMBER PREFIX	PAVER MODEL	Tamper Bar	Tamper Heating Element
AS4173	NET	AP300	515-7602*	
AS4251 Electric	DLH	AP600, AP755	475-6263*	475-6284*
AS4251C Diesel	DLB	AP500E, AP555E, AP600D, AP655D	475-6259*	
AS4251C Electric	DLB	AP500E, AP555E, AP600D, AP655D	475-6263*	488-3928*
AS4252C Electric	TPS	AP500E, AP555E, AP600D, AP655D	475-6285*	475-6289
SE34 VT	NEK, NEN	AP300, AP300F, AP355, AP355F	515-1561	515-1560
SE47 VT	TYR	AP400, AP455, AP500, AP555	626-4381*	626-4382*
SE50 VT	A42	AP500, AP555	626-4383*	626-4384*
	E5T	AP500F, AP555F, AP600, AP600F, AP655, AP655F, AP655L, AP655F L	574-8019*	578-9455*
	B86	AP500F, AP555F, AP600F, AP655F, AP655F L	574-8018*	578-9455*
	J6K		Serial Number J6K00181 and up: 574-8019*	
SE60 VT XW	XW6	AP600, AP655, AP655L, AP1000, AP1055	574-8017*	578-9454*
	E99, TLW	AP655F, AP655F L, AP1000F, AP1055F	574-8015*	
			Serial Number TLW00220 and up: 574-8017*	

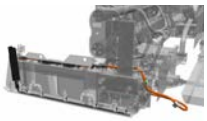
** This is a paper kit. Not available as a stocked kit but as a service repair option that may be ordered, at the consist level, through your Cat dealer to complete the repair or use SIS to identify part numbers for order.*



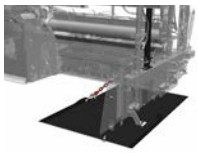


END GATE DEPTH ADJUSTMENT Standard manually adjustable with crank jacks or can be upgraded to hydraulically powered option.

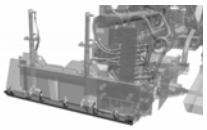
DEEP DEPTH END GATES Manual end gates with 380 mm (15 in) travel for deep lift paving—75 mm (3 in) above grade and 305 mm (12 in) below grade. Replaces the end gates that ship with the machine. Can be used with folding end gates, but cannot be used with the heated or powered end gate options.



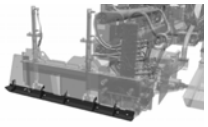
HEATED END GATES Includes two heating elements to heat the end gate shoes. For folding or non-folding end gates.



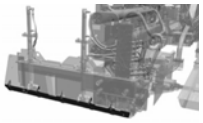
CUTOFF SHOE Reduces the paving width to a measurement less than the width of the main screed in 152 mm (6 in) increments.



STANDARD END GATE SHOE Wearing portion of the end gate that rides on the ground and is 50 mm (2 in) wide. Shoe can be used with the end gate heater.



WIDE END GATE SHOES These shoes are wider than the end gate structure at 88 mm (3.5 in) wide. Shoe can be used with the end gate heater.



EDGE SHOES Vertical end gate shoes can replace standard end gate shoes. Mounting tabs are welded to the outside of the vertical structure. The vertical shoe is 5 mm (0.2 in) thick and has 16 mm (0.6 in) of clearance from the bottom wear surface up to the mounting tabs. Total width at the mounting tabs is 50 mm (2 in). The vertical end gate shoes do not work with end gate heaters.



SAFETY EDGE SHOE Reversible safety edge shoe for use on either side of the screed. Attaches to the rear of the end gate and modifies the edge of the mat as it leaves the screed. Mounts to the same location as the folding end gate and cannot be used while the folding end gate is installed.

POWER FOR HYDRAULIC END GATES Provides hydraulic motor power for raising and lowering extender height relative to the main screed. Also provides hydraulic power for powered end gates.

END GATE AND SHOE OPTIONS

DESCRIPTION		MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER
Available for order through your Cat dealer only.				
Adjustable End Gate Crank	Includes four jacks	SE47 FM	4B5	499-1424
		SE47 FM, SE60 FM	411, FM6	247-7616
		SE47 V, SE47 VT, SE50 VT, SE60 VT XW	3M6, A42, E5T, TYR, XW6	528-8322*
		SE50 V, SE50 VT, SE60 V	B86, J6K, J97, JJ3	538-9613*
		SE60 V XW, SE60 VT XW	E99, JS9, TLW	528-8321*
Hydraulic End Gate Cylinders and Switch	Left and right	SE47 V	3M6	599-0729*
		SE47 VT	TYR	602-3908*
		SE50 VT	A42	602-3443*
		SE50 V, SE50 VT, SE60 V	B86, E5T, J6K, J97, JJ3	523-4576*
		SE60 V XW, SE60 VT XW	E99, JS9, TLW, XW6	523-4577*
Heating Elements for Non-Folding End Gate Shoes	Left and right	SE34 V, SE34 VT	NEK, RAT	449-0360*
		SE47 FM	4B5	571-3783*
		SE47 FM, SE60 FM	411, FM6	563-8391*
		SE47 V, SE47 VT, SE50 VT	3M6, A42, TYR	570-0431*
		SE50 V, SE50 VT, SE60 V	B86, J6K, J97, JJ3	490-1681*
		SE60 V XW, SE60 VT XW	E99, JS9, TLW, XW6	490-1690*
Heating Elements for Folding End Gate Shoes	Left and right	SE50 VT	A42	570-0431*
		SE50 V, SE50 VT	B86, E5T, J6K, J97	438-1030*
		SE60 V XW, SE60 VT XW	E99, JS9, TLW	446-5919*
Deep Depth End Gates	Left and right	SE34 VT	NEK	485-4030*
		SE50 VT, SE60 VT XW	B86, E99, J6K, TLW	453-2460*
Cutoff Shoe	One shoe	SE47 FM, SE60 FM	411, 4B5, FM6	328-3562*
		SE47 V, SE47 VT, SE50 VT	3M6, A42, TYR	570-0429*
		SE50 V, SE50 VT, SE60 V	B86, E5T, J6K, J97, JJ3	490-2550*
		SE60 V XW	JS9	490-2560*
		SE60 VT XW	E99, TLW, XW6	490-2570*
Standard End Gate Shoe	Left and right	SE47 FM, SE60 FM	411, 4B5, FM6	630-4884
		SE47 V, SE47 VT, SE50 VT	3M6, A42, TYR	610-4623
	Left	SE50 V, SE50 VT, SE60 V	B86, E5T, J6K, J97, JJ3	499-3591
		SE60 V XW, SE60 VT XW	E99, JS9, TLW, XW6	
	Right	SE50 V, SE50 VT, SE60 V	B86, E5T, J6K, J97, JJ3	499-3590
		SE60 V XW, SE60 VT XW	E99, JS9, TLW, XW6	

** This is a paper kit. Not available as a stocked kit but as a service repair option that may be ordered, at the consist level, through your Cat dealer to complete the repair or use SIS to identify part numbers for order.*

Continued

END GATE AND SHOE OPTIONS

DESCRIPTION		MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER
These are paper kits and not available as a stocked kit but as a service repair option that may be ordered, at the consist level, through your Cat dealer to complete the repair or use SIS to identify part numbers for order. *Available for order through your Cat dealer only.*				
Wide End Gate Shoes	Two shoes	SE47 V, SE47 VT, SE50 VT	3M6, A42, TYR	604-6340
		SE50 V, SE50 VT, SE60 V	B86, E5T, J6K, J97, JJ3	472-0510
		SE60 V XW, SE60 VT XW	E99, JS9, TLW, XW6	
Edge Shoes	Left and right	SE47 V, SE47 VT, SE50 VT	3M6, A42, TYR	570-0430
		SE50 V, SE50 VT, SE60 V	B86, E5T, J6K, J97, JJ3	472-0512
		SE60 V XW, SE60 VT XW	E99, JS9, TLW, XW6	
Safety Edge Shoe *63.5 mm (2.5 in)*	One shoe	SE47 FM, SE60 FM	411, 4B5, FM6	580-7894
		SE47 V, SE47 VT, SE50 VT	3M6, A42, TYR	586-8321
Safety Edge Shoe *127 mm (5 in)*	One shoe	SE47 FM, SE60 FM	411, 4B5, FM6	587-8960
		SE47 V, SE47 VT, SE50 VT	3M6, A42, TYR	570-0432
		SE50 V, SE50 VT, SE60 V	B86, E5T, J6K, J97, JJ3	467-4575
		SE60 V XW, SE60 VT XW	E99, JS9, TLW, XW6	
Actuator Safety Edge		SE47 FM, SE47 V, SE50 VT	3M6, 4B5, A42	610-3591
		SE47 FM, SE60 FM	411, FM6	627-5406
		SE47 VT	TYR	570-0429
Power for Hydraulic End Gates and/or Extender Height		SE50 V	J97	473-3320
		SE60 V	JJ3	473-3321
Berm	12 inch	SE47 FM	4B5	571-3762
			411	552-8039
		SE60 FM	FM6	600-1666
	18 inch	SE47 FM	4B5	571-3777
			411	552-8040
		SE60 FM	FM6	600-1669
	24 inch	SE60 FM	FM6	600-8321
Berm Prep		SE47 FM	4B5	590-1814
			411	590-1703
		SE60 FM	FM6	600-1672

** This is a paper kit. Not available as a stocked kit but as a service repair option that may be ordered, at the consist level, through your Cat dealer to complete the repair or use SIS to identify part numbers for order.*

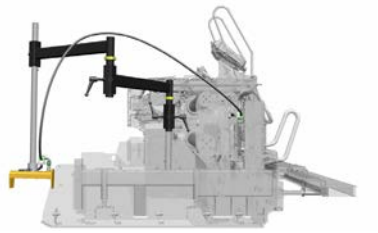


GRADE CONTROL

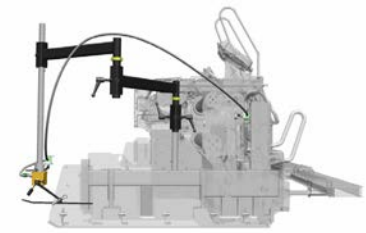
PRECISE AND REPEATABLE SMOOTHNESS

The Cat GRADE with Grade and Slope system for pavers removes irregularities from the paved surface. The system consists of sensors to control elevation and slope. These sensors can work individually or together to make automatic elevation or cross slope adjustments and maintain the targeted values. Averaging beams equipped with multiple sensors provide a longer averaging area to help remove irregularities for smoother transitions.

SENSORS



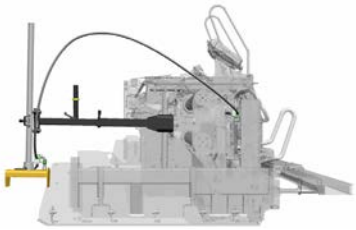
SONIC GRADE SENSOR
One sonic grade sensor, coily cord, sensor pole, two support arms, and mounting hardware. Two sensors are required for basic operation. Three sensors are required for use with the Sonic Averaging Beam.



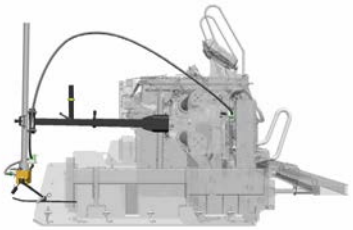
MECHANICAL GRADE SENSOR
One proportional contacting mechanical sensor, wand, riding shoe, coily cord, sensor pole, two support arms, and mounting hardware Two sensors are required for basic operation. Three sensors are required for use with the Sonic Averaging Beam.



SLOPE SENSOR
One slope sensor with hardware to be mounted to the transverse beam.



EXTENDABLE SONIC GRADE SENSOR
One sonic grade sensor, coily cord, sensor pole, telescoping sensor mount, and mounting hardware. Two sensors are required for basic operation. Three sensors are required for use with the Sonic Averaging Beam.



EXTENDABLE MECHANICAL GRADE SENSOR
One proportional contacting mechanical sensor, wand, riding shoe, coily cord, sensor pole, telescoping sensor mount, and mounting hardware. Two sensors are required for basic operation. Three sensors are required for use with the Sonic Averaging Beam.

GRADE SENSORS

DESCRIPTION	MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER	QUANTITY
Available for order through your Cat dealer only.				
Cat Grade and Slope Controller *(Vibratory)*	AP400, AP455	599, 676, A45, MN6, P45, R4N	634-3524	1
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		
	AP600, AP655, AP655 L	3A6, 668, 671, 695, 696, 697, 6T5, 9P7, A66, F67, N66, P7C, R66		
	AP1000, AP1055	698, 699, F7T		
Cat Grade and Slope Controller *(Tamper)*	AP400, AP455	599, 676, A45, MN6, P45, R4N	590-1429	1
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		
	AP600, AP655, AP655 L	3A6, 668, 671, 695, 696, 697, 6T5, 9P7, A66, F67, N66, P7C, R66		
	AP1000, AP1055	698, 699, F7T		
Sonic Grade Sensor	AP300, AP355 AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	449-0383	1*
	AP400, AP455	599, 676, A45, MN6, P45, R4N	561-8022	
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		
	AP600, AP655, AP655 L	3A6, 668, 671, 695, 696, 697, 6T5, 9P7, A66, F67, N66, P7C, R66	602-9481	
	AP1000, AP1055	698, 699, F7T		
	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	416-2006	
	AP600F, AP655F, AP655F L	452, 453, 479, 480, 655, AC6, AP6, D6F, JA6, LR6, MH6, ML6, P65		
	AP1000F, AP1055F	AC4, RLM, TJ5		
Extendable Sonic Grade Sensor *(includes mounts)*	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	459-6770	1*
	AP600, AP655, AP655 L	3A6, 668, 671, 695, 696, 697, 6T5, 9P7, A66, F67, N66, P7C, R66	602-9483	
	AP1000, AP1055	698, 699, F7T		
	AP600F, AP655F, AP655F L	452, 453, 479, 480, 655, AC6, AP6, D6F, JA6, LR6, MH6, ML6, P65	459-6770	
	AP1000F, AP1055F	AC4, RLM, TJ5		
Extendable Sonic Grade Sensor Mount	AP400, AP455	599, 676, A45, MN6, P45, R4N	599-2631	1*
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		

** For basic operation, two sensors are required. If using with the Sonic Averaging Beam, three sensors are required.*

Pavers require the Cat Grade and Slope System to be installed. If your paver is not equipped with the Cat Grade and Slope system, it can be ordered from your SITECH dealer. (Part Number 416-2003).

Continued

GRADE SENSORS

DESCRIPTION	MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER	QUANTITY
Available for order through your Cat dealer only.				
Mechanical Grade Sensor *(sensor only)*	AP400, AP455	599, 676, A45, MN6, P45, R4N	561-8023	1
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		
Mechanical Grade Sensor *(includes mounts)*	AP300, AP355, AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	449-0381	1*
	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	416-2007	1*
	AP600, AP655, AP655 L	3A6, 668, 671, 695, 696, 697, 6T5, 9P7, A66, F67, N66, P7C, R66	602-9482	
	AP1000, AP1055	698, 699, F7T		
	AP600F, AP655F, AP655F L	452, 453, 479, 480, 655, AC6, AP6, D6F, JA6, LR6, MH6, ML6, P65	416-2007	
	AP1000F, AP1055F	AC4, RLM, TJ5		
Extendable Mechanical Grade Sensor Mount	AP400, AP455	599, 676, A45, MN6, P45, R4N	599-2631	1
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		
Extendable Mechanical Grade Sensor *(includes mounts)*	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	513-0333	1
	AP600, AP655, AP655 L	3A6, 668, 671, 695, 696, 697, 6T5, 9P7, A66, F67, N66, P7C, R66	602-9484	
	AP1000, AP1055	698, 699, F7T		
	AP600F, AP655F, AP655F L	452, 453, 479, 480, 655, AC6, AP6, D6F, JA6, LR6, MH6, ML6, P65	513-0333	
	AP1000F, AP1055F	AC4, RLM, TJ5		
Adjustable Z-Style Sensor Mount	AP400, AP455	599, 676, A45, MN6, P45, R4N	599-2630	1
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		
Slope Sensor	AP300, AP355, AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	449-0382	1
	AP400, AP455	599, 676, A45, MN6, P45, R4N	561-8024	1
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678		
	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	362-8687	1
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66		
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5		

** For basic operation, two sensors are required. If using with the Sonic Averaging Beam, three sensors are required.*

Pavers require the Cat Grade and Slope System to be installed. If your paver is not equipped with the Cat Grade and Slope system, it can be ordered from your SITECH dealer. (Part Number 416-2003).

BEAMS AND LEVELERS



SONIC AVERAGING BEAM
A 9.1 m (30 ft) non-contacting beam to be used with three sonic sensors for enhanced averaging capability.



FORE 'N AFT LEVELER
A 3.7 m (12 ft) multi-pad aluminum beam that mounts ahead and inboard of the screed with an additional towed shoe attached to the screed that rides on the mat.



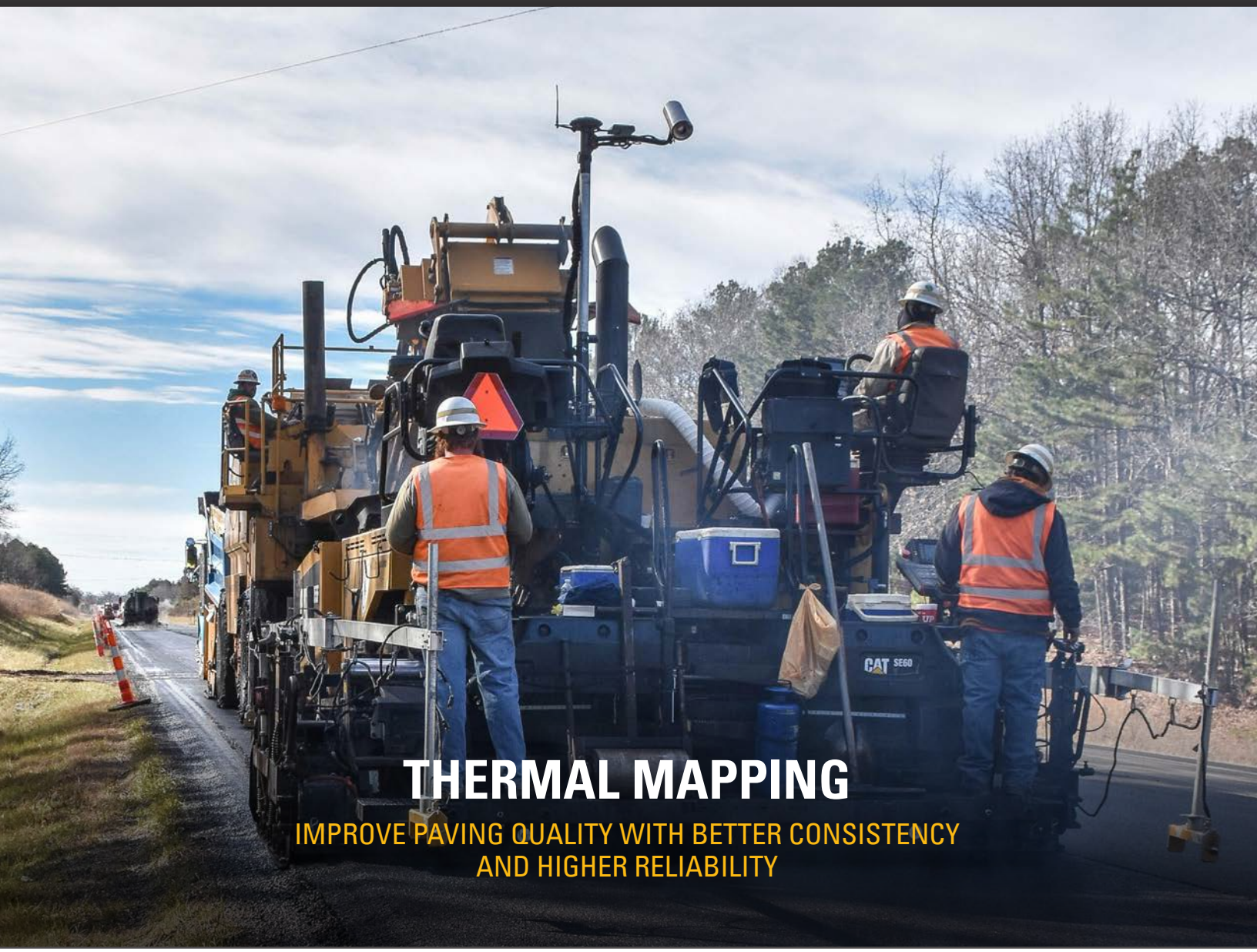
9.1 M (30 FT) OUTBOARD LEVELER
A 9.1 m (30 ft) long multi pad aluminum beam that runs along the outside of the tractor and the screed.

12.2 M (40 FT) OUTBOARD LEVELER
A 12.2 m (40 ft) long multi pad aluminum beam that runs along the outside of the tractor and the screed.

GRADE AND SLOPE BEAMS AND LEVELERS

DESCRIPTION	MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER	QUANTITY
Available for order through your Cat dealer only.				
Sonic Averaging Beam	AP400, AP455	599, 676, A45, MN6, P45, R4N	417-1907	1*
	AP500, AP555, AP500F, AP555F	449, 4M6, 503, 5F5, 5P5, 636, 637, 639, 678, AF5, AP5, F5M, M45, RBM		
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66		
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5		
Fore 'N Aft Leveler	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	109-5625	1
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66		
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5		
9.1 m (30 ft) Outboard Leveler	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	109-5623	1
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66		
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5		
12.2 m (40 ft) Outboard Leveler	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	109-5622	1
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66		
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5		

** Three sonic sensors are required for operation.*



THERMAL MAPPING

IMPROVE PAVING QUALITY WITH BETTER CONSISTENCY AND HIGHER RELIABILITY

Thermal segregation is one of the leading causes of road failure in the asphalt paving industry. Viewing real-time temperatures allows contractors to identify variations and take action to manage the plant to paver delivery process and fine-tune paving practices for more uniform lay-down temperatures to increase consistency up to 15%.

TAKE CONTROL OF THE OUTCOME

Early detection of thermal variation enables timely process control adjustments to be made allowing you to:

MEET PERFORMANCE SPECIFICATIONS

+ Meet AASHTO thermal mapping specifications
+ Optimize compaction for PWL Specifications
+ Earn more incentive money by exceeding pay for performance quality targets

DETECT MAT TEMPERATURE VARIATION

+ Monitor temperatures in real-time and make necessary adjustments
+ Help ensure mix being delivered from the plant has a uniform temperature
+ Fine-tune paving practices to help ensure consistent laydown techniques

GET RELIABLE PERFORMANCE AND SUPPORT

+ System consists of no moving parts
+ Camera mounts on fixed-position mast
+ Localized SITECH support



HOW IT WORKS

Thermal mapping utilizes an infra-red camera mounted on a robust, fixed-position mast. The camera captures the entire width of the asphalt surface behind the screed. An LCD display mounted near the screed console provides real-time visibility to the surface being monitored. Data is automatically recorded and stored for future analysis.

Both on-board and off-board storage is available. The live view helps ensure that the mix temperature is uniform and being placed at the designed temperature. It can also alert operators of potential issues before they become a major rework project. An in-field report can be generated that assesses the overall temperature variation.

Thermal mapping utilizes an infra-red camera, GNSS antenna, fixed-position mast and an LCD Display. Simple set-up helps contractors get the most from their investment. The Cat Thermal Mapping Kit includes certain components and installation hardware. Additional items need to be sourced from your local SITECH dealer. ***Thermal Mapping cannot be used on pavers equipped with a hard top canopy.***



THERMAL CAMERA AND MS955 ANTENNA
The camera measures surface temperature of the asphalt layer and the next generation antenna establishes machine position and thermal mapping geospatial location.



SNM941 MODEM
The modem is used to off-board thermal map directly into VETA and is capable of Wi-Fi communication (machine-to-machine thermal sharing).



CB460 DISPLAY
An LCD screen displays the thermal map to screed users and is capable of in-field reports for thermal variation.

THERMAL MAPPING KIT			
DESCRIPTION	**MACHINE MODEL**	**SERIAL NUMBER PREFIX**	**PART NUMBER**
Available for order through your Cat dealer only. Additional items must be sourced from your SITECH dealer. ***Thermal Mapping may not be available in all regions. Please consult your Cat dealer for availability in your location.***			
Contents include: • Camera mounting mast • Thermal camera • MS955 receiver/antenna • Mounting provisions for the CB460 display • Electronics housing • Wiring for SITECH supplied items	AP500, AP555, AP500F, AP555F	449, 4M6, 503, 5F5, 5P5, 636, 637, 639, 678, AF5, AP5, F5M, M45, RBM	526-9342
	AP600, AP655, AP600F, AP655F	3A6, 452, 453, 479, 480, 655, 668, 695, 696, 697, 6T5, 9P7, A66, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	

ADDITIONAL ITEMS SOURCED FROM YOUR SITECH DEALER

MANDATORY:

+ SNM941 modem for off-boarding of data (requires 3rd party cellular data plan to send data)
+ RTK position accuracy from either a local base station or Virtual Reference Station (VRS) correction source
+ SNR radio, GNSS upgrade keys, Trimble Connected Community license
+ Thermal mapping only or 3D and thermal mapping software key for the CB460 display

OPTIONAL:

+ CB460 display (If also using 3D, an additional screen is needed. The CB460 display cannot be used simultaneously for 3D and thermal mapping)
+ If not installing on AP500, AP500F, AP555, AP555F, AP600, AP600F, AP655, AP655F, AP1000, AP1000F, AP1055 or AP1055F pavers, an adapter plate is needed for mounting

Pavers require the Cat Grade and Slope System to be installed. If your paver is not equipped with the Cat Grade and Slope system, it can be ordered from your SITECH dealer. (Part Number 416-2003).

In order for thermal mapping to work as expected, Paver Mounted Thermal Profiling (PMTP) mapping must be created at RTK accuracy. Failure of the RTK signal causes unusable positioning data.



TRACTOR AND SCREED OPTIONS

UPGRADES TO INCREASE CAPABILITY

Start with the best in efficiency and smoothness—and then customize it with upgrades that enhance our most popular features. From material delivery to screed accessories and everything in between, your asphalt paver and screed can be tailored to specific operating conditions or preferences.

SONIC MATERIAL FEED SENSORS

Four sonic transducers mounted above the conveyor discharge and at the end of the augers activate automatically to maintain a preset amount of material ahead of the screed.

Two sonic transducers mounted at the end of the augers automatically maintain a preset amount of material ahead of the screed.



PADDLE MATERIAL FEED SENSORS

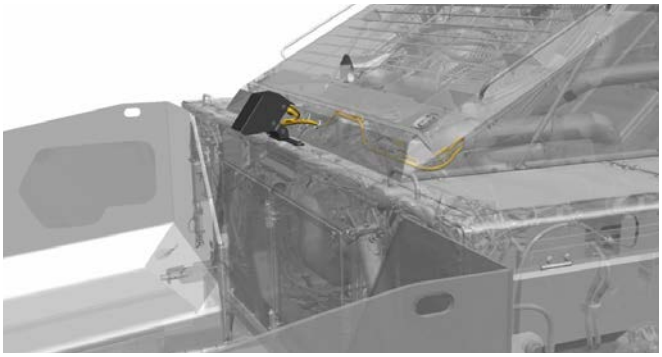
Two paddle sensors located in the auger chamber activate the conveyors to automatically maintain the desired amount of material ahead of the screed.



HOPPER MATERIAL LEVEL AND TEMPERATURE SENSOR

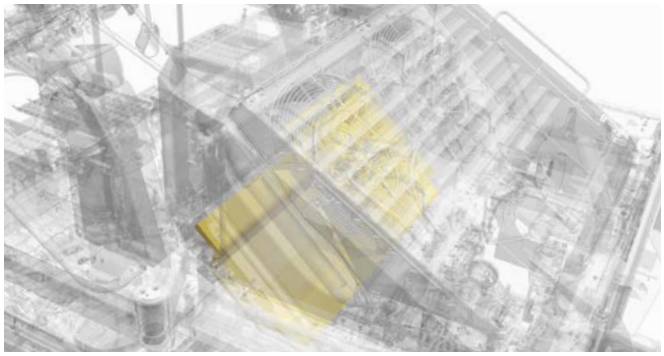
Mounts above the hopper to monitor the amount of material and the temperature of the mix using a single infrared temperature sensor. Material level and temperature information is available on the machine display at the operator station.

Not compatible with certain serial numbers. Before ordering, provide machine serial number to your Cat dealer.



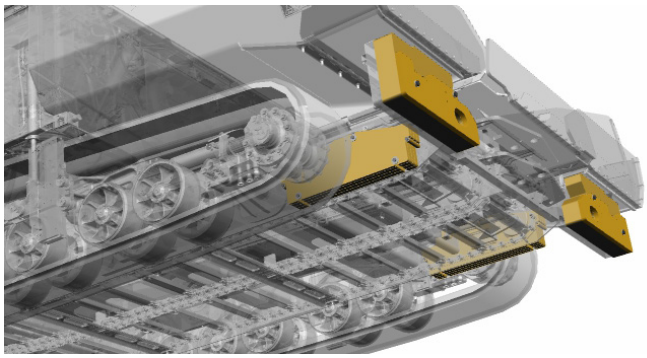
COOLING PACKAGE FILTER

Fiberglass filters and installation rod to protect the cooling package from build up of asphalt polymers.



BALLASTS

Additional weight for wide width paving helps balance the weight of the screed. Different packages are available to match your application.



TOW POINT INDICATOR

Pair of tow point height gauges that mount on top of paver deck. The upper tow-point indicator is mounted to the top deck near the hopper, providing a higher visual reference of tow-point height.



STEERING GUIDE

Adjustable position steering guide with indicator chain. The steering guide attaches to the front of the tractor. A chain hangs off the end of the shaft in order to help the operator follow a line or positioning device. Quantity: 1



HARDTOP CANOPY

The hardtop canopy offers protection from the elements for the operator. Extendable wings provide additional protection and when operator station is swung out.

Exhaust stack will need to be modified when the canopy option is installed on select models. Contact your Cat dealer for additional hardware to make this modification.



WINDSHIELD

Dual windshields with adjustable wipers are available on pavers equipped with a hardtop canopy and provide operator comfort. The windshields are retractable for additional versatility.



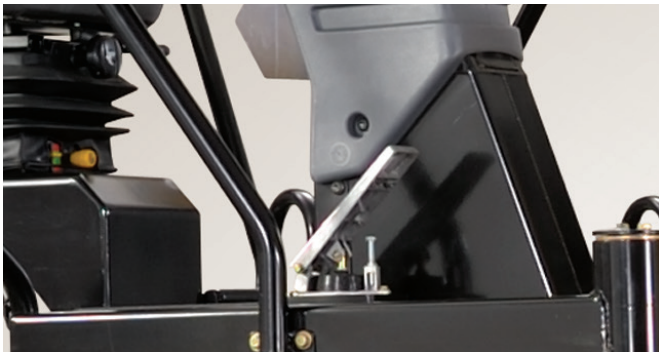
UMBRELLA

Provides paver and screed operators protection in adverse weather conditions. Umbrellas measure approximately 140 x 140 cm (54 x 54 in) and are collapsible. Includes mounting hardware. Umbrellas available for tractor or screed.



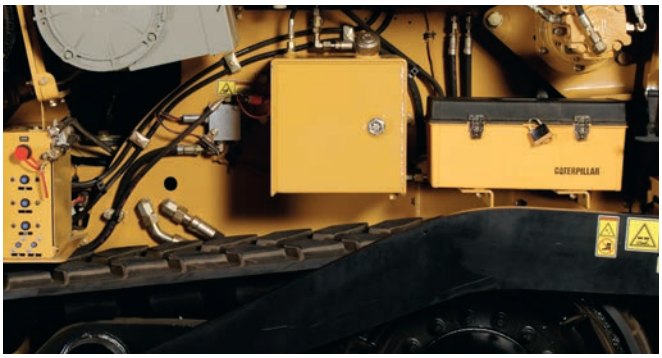
DECEL PEDAL

The foot-operated pedal provides deceleration as well as dynamic braking without decreasing the propel speed setting. Includes a quantity of two, one for each operator station.



ECOLOGICAL WASHDOWN TANK

A 30 L (8 gal) tank with integral pump mounted of the tractor. Intended for use with ecological cleaning fluids. Pump is included.



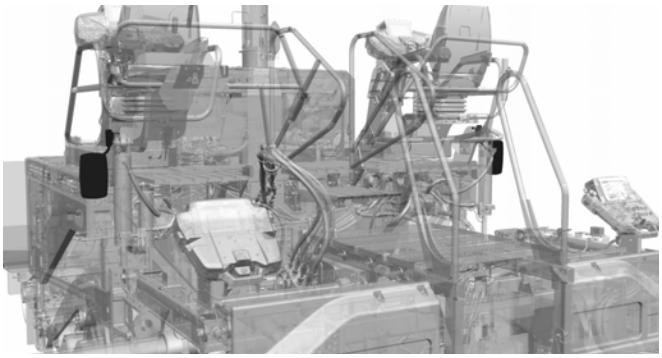
HOSE REEL WITH WAND

Includes 9 m (30 ft) hose on self-storing reel with spray wand mounted to left-hand handrail at the rear of the tractor.



MIRRORS

Includes two convex mirrors mounted outboard of the screed lift cylinders to provide the screed operator with a clear view of material in front of the screed extensions and to see traffic or equipment behind them.



DISPLAY VISORS

Includes two folding visors that shade the displays from sunlight. The visors fold back to allow the installation of the vandal covers. Screed and tractor visors need to be ordered separately.



SCREED CONTROL PENDANT

Allows adjustment of optional powered endgate, screed width, extension height and slope and manual override and pause of the feeders system. Includes two removable pendant controls with coily cord.



SCREED SWIVEL CONTROL MOUNTS

Includes a left and right mounting bracket to allow the left and right screed operator controls to swivel up/down and left/right.



MATERIAL FEED SENSORS

DESCRIPTION	MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER
Available for order through your Cat dealer only.			
Two Sonic Sensors *(augers)*	AP300, AP355, AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	478-3977
	AP400, AP455	599, 676, A45, MN6, P45, R4N	575-0376
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678	
	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	416-1467
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	
Two Sonic Sensors *(conveyors)*	AP400, AP455	599, 676, A45, MN6, P45, R4N	575-0377
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678	
Four Sonic Sensors *(augers / conveyor)*	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	465-1275
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	33A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	
Two Paddle Sensors *(augers)*	AP300, AP355, AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	608-1427
	AP400, AP455	599, 676, A45, MN6, P45, R4N	560-2289
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678	
	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	416-1465
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	
Two Paddle Sensors *(conveyors)*	AP400, AP455	599, 676, A45, MN6, P45, R4N	560-2297
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678	
Four Paddle Sensors *(augers / conveyor)*	AP300, AP355, AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	608-1428

Continued

MATERIAL FEED SENSORS

DESCRIPTION	MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER
Available for order through your Cat dealer only.			
Two Paddle, Two Sonic Sensors *(augers / conveyor)*	AP300, AP355, AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	608-1429
	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	435-0910
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	33A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	416-1468
Hopper Material Level and Temperature Sensor*	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	574-5683
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	33A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	510-5990
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	510-5974

** Not compatible with certain serial numbers. Before ordering, provide machine serial number to your Cat dealer.*

FILTER COOLING PACKAGE

DESCRIPTION	MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER
Available for order through your Cat dealer only.			
Filter Cooling Package	AP400, AP455	599, 676, A45, MN6, P45, R4N	568-6436
	AP500, AP555	4M6, 5P5, 636, 637, 639, 678	
	AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	33A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	572-2979
	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	572-2982

BALLASTS

DESCRIPTION	WEIGHT	MACHINE MODEL	SERIAL NUMBER PREFIX	PART NUMBER
Available for order through your Cat dealer only.				
2-Plate Bumper	151 kg (333 lbs)	AP400	599, R4N	585-3208
		AP500, AP555	4M6, 5P5, 636, 637, 639, 678	585-1935
		AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	435-1041
		AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	179 kg (395 lbs)	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	416-4107
4-Plate Bumper	303 kg (668 lbs)	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	435-1042
		AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	357 kg (787 lbs)	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	431-0691
6-Plate Bumper	352 kg (777 lbs)	AP500F, AP555F	449, 503, 5F5, AF5, AP5, F5M, M45, RBM	435-1043
		AP600, AP655, AP655 L, AP600F, AP655F, AP655F L	3A6, 452, 453, 479, 480, 655, 668, 671, 695, 696, 697, 6T5, 9P7, A66, AC6, AP6, D6F, F67, JA6, LR6, MH6, ML6, N66, P7C, P65, R66	
	413 kg (910 lbs)	AP1000, AP1055, AP1000F, AP1055F	698, 699, AC4, F7T, RLM, TJ5	431-0692
8-Plate Bumper	418 kg (921 lbs)	AP300, AP355, AP300F, AP355F	7P3, 7P5, 7P7, 7P8, 7P9, 7PF, FRA, END, NED, NEE, NRG, NRH, NNN, NNP, NTT, NTW, WH2, WHY, YES, YET	488-6301
Side Plate	133 kg (293 lbs)	AP655, AP655F	452, 453, 479, 480, 655, 668, 696, 697, 6T5, 9P7, F67, MH6, ML6, P65, P7C, R66	435-1044
	243 kg (535 lbs)	AP655 L, AP655F L	671, AC6	522-6378
	337 kg (743 lbs)	AP1055, AP1055F	699, F7T, RLM, TJ5	431-0693